

VL 402: RFIC Design

Course Code: VL 402

Course Name: RFIC Design

L-T-P-C: 3-0-0-3

Prerequisites: VL 211: Analog Circuit Design and EE 202: Electromagnetic and Wave Propagation

Intended for: Discipline Core for B.Tech in VLSI and Microelectronics Engineering Students.

Mutual Exclusion: None

Approval: 61st BoA

Preamble

In today's rapidly evolving world, wireless communication is transforming industries and daily life. From Bluetooth devices and video game controllers to GPS navigation and satellite TV, radio frequency (RF) integrated technology powers it all. As RF Engineering reshapes global connectivity, there has never been a more exciting time to master this field with the essential skills to excel in both the commercial and military sectors. Learn how to design cutting-edge on-chip RF circuits that are energy-efficient, high-performance, and cost-effective, enabling the next generation of wireless technology. This program equips you with hands-on knowledge to create RF subsystems, design integrated circuits, and master the fundamentals of RF systems. By the end of this program, you will: a) Understand the unique design requirements at RF frequencies, b) Differentiate between RF circuit, integrated circuit (IC), and system design needs, c) Master the design of RF blocks and systems, including calculating system characteristics.

Course Modules with Quantitative Lectures Hours

- **I. Microwave Network Representations:** ABCD parameters, Scattering parameters, Relationship with other representations and General block diagram of RFIC chip. (3 hours)
- **II. Planar Transmission Line:** Microstrip transmission line in layered medium, Analysis using variational techniques, Strip transmission line and their closed form expression, Analysis of Multiconductor transmission line to identify coupling properties at RF frequency. (6 hours)
- **III. Lumped Elements Integrated Circuit Technology in RF Frequency:** Design of lumped elements (resistor, inductor, and capacitor) and measurement of their parameters. (3 hours)
- **IV. Bipolar/MOS Models for Integrated Active Device:** Large signal model (BJT), Small signal model (BJT). (2 hours)
- **V. Integrated Circuit Technology:** Basic process in integrated circuit, Bipolar integrated circuit fabrication, Bi-CMOS technology. (2 hours)
- **VI. Transistors Circuit Topology:** Current source, Different bias circuits and active load, Two transistor amplifier stages (Emitter coupled pair), Emitter follower as an output stage, Analog multiplier using bipolar transistors, Phase lock loop, Multi-tanh (hyperbolic) circuit. (6 hours)
- **VII. Basic Concept of RF Design:** Nonlinearity and time variance, Inter symbol interference, Characterization of different noises and its measurement techniques, Sensitivity and Dynamic range, Passive impedance transformation. (7 hours)

- **VIII. Low Noise Amplifier and Mixer:** General consideration, Bipolar LNA and input matching, Down conversion mixer; General consideration, Bipolar mixer, Cascaded stage revisited. (5 hours)
- **IX. Oscillator:** LC oscillator topology, Phase noise, Bipolar oscillator. (4 hours)
- **X. Power Amplifier:** Class A/B amplification, Class C amplifier, Biasing techniques, Class D, E, and F amplifier. (4 hours)

Text Books

1. *Analysis and Design of Analog Integrated Circuits*: P. R. Gray, and R. G. Meyer.
2. *RF Microelectronics*: B. Razavi.

References

1. *Computer-aided design of microwave circuits*: K.C. Gupta.
2. *Solid State Radio Engineering*: Herbert L. Krauss